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Error correction methods for computational SSD supporting rapid file semantic search

Abstract

Devices and methods to implement semantic searching on SSD through a computational SSD system that distributes computing to each NAND flash die of the SSD while the SSD controller handles the results aggregation with new on-die computation logic circuits to provide on device file semantic search are disclosed herein. The computational SSD system can read file feature vectors from multiple dies to the SSD controller, and if needed, these feature vectors may be buffered in DRAM and controller handles distance computing. Local, on-die AI/ML processing units may perform, for example, computation and comparison operations and pass the processing scores and results to the SSD controller. The SSD controller aggregates results from all dies and returns the result to the host. The feature vector store size, circuitry and number of on-die AI/ML processing units may be configured as needed to adapt to different tasks, system constraints, and/or feature vector sizes.

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Background/Summary

PRIORITY (1) This application claims the benefit of and priority to U.S. Provisional Application No. 63/476,666, filed Dec. 22, 2022, which is incorporated in its entirety herein.

FIELD

(1) The present disclosure relates to storage systems. More particularly, the present disclosure relates to utilizing storage systems to perform machine learning processes.

BACKGROUND

(2) Storage devices are ubiquitous within computing systems. Recently, solid-state storage devices (SSDs) have become increasingly common. These nonvolatile storage devices can communicate and utilize various protocols including non-volatile memory express (NVMe), and peripheral component interconnect express (PCIe) to reduce processing overhead and increase efficiency.

(3) As they have evolved, SSDs have become more power efficient compared to traditional hard disk drives (HDDs), hence giving SSDs an advantage in consumer and commercial markets. Over the years, many users have accumulated thousands of files documenting their lives, experiences, travels, and interests. To find particular media files based on their content, users often spend many hours navigating, organizing, and searching albums and folders for specific photos, videos, and other media. Generally, user media is stored across various storage and computing devices, and users have limited methods for contextually searching photos, videos, and other media to find the desired media file. Indeed, users are limited to conventional search methods such as lexical search for finding media on storage systems which are limited to searching for recent files or files with specific filenames or attributes. Moreover, the sheer volume of stored media files and lack of contextual search methods make it difficult for users to find specific media on their storage and computing devices.

(4) With the development of Artificial Intelligence (AI) and Machine Learning (ML), accurate semantic searching has become possible allowing users to find relevant content within their media files stored on their storage devices. However, conventional implementations of AI/ML semantic searching of storage devices have some problems that make it inefficient and limited. One problem in conventional AI/ML implementations of semantic searching is that data is repeatedly moved between the host processor, the host memory, and the storage device leading to substantial waste of bandwidth and energy from data movement. Another problem in conventional AI/ML implementations of semantic searching is in preparing data for AI/ML processing for on host computing, that is, reading data from multiple dies to the storage device controller and buffering the data in memory for the controller to handle AI/ML processing, then performing AI/ML processing. This causes a bottleneck for the storage device controller, as the controller has to handle both I/O and computing operations, and such computation requires additional memory buffer to store data for AI/ML processing.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above, and other, aspects, features, and advantages of several embodiments of the present disclosure will be more apparent from the following description as presented in conjunction with the following several figures of the drawings.

(2) FIG. 1 is a schematic block diagram of an example host-computing device with a storage system suitable for computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(3) FIG. 2 is a schematic block diagram of an example storage device suitable for computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(4) FIG. 3 is a conceptual illustration of a page of memory cells, organized for example in the NAND configuration, being sensed or programmed in parallel in accordance with an embodiment of the disclosure;

(5) FIG. 4 is a schematic block diagram of an example computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(6) FIG. 5 is a schematic block diagram of an example machine learning processing unit of the computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(7) FIG. 6 is a schematic block diagram of an example machine learning processing unit of the computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(8) FIG. 7 is a flowchart depicting a process for utilizing an example computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(9) FIG. 8 is a flowchart depicting a process for utilizing an example computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(10) FIG. 9 is a schematic block diagram of an example machine learning process in accordance with an embodiment of the disclosure;

(11) FIG. 10A is a schematic block diagram of an example error correction system in accordance with an embodiment of the disclosure;

(12) FIG. 10B is a chart of the example error correction system of FIG. 10A in accordance with an embodiment of the disclosure;

(13) FIG. 10C is a table of the example error correction system of FIG. 10A in accordance with an embodiment of the disclosure;

(14) FIG. 11A is a schematic block diagram of an example error correction system in accordance with an embodiment of the disclosure;

(15) FIG. 11B is a chart of the example error correction system of FIG. 11A in accordance with an embodiment of the disclosure;

(16) FIG. 11C is a table of the example error correction system of FIG. 11A in accordance with an embodiment of the disclosure;

(17) FIG. 12 is a schematic block diagram of an example error correction mode and storage mode of the computational SSD system supporting semantic search with error correction systems in accordance with an embodiment of the disclosure; and

(18) FIG. 13 is a schematic block diagram of another example error correction mode and storage mode of the computational SSD system supporting semantic search with error correction systems in accordance with an embodiment of the disclosure.

(19) FIG. 14 is a flowchart depicting a process for utilizing an example error correction method in a computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(20) FIG. 15 is a flowchart depicting a process for utilizing an example error correction method in a computational SSD system supporting semantic search in accordance with an embodiment of the disclosure;

(21) Corresponding reference characters indicate corresponding components throughout the several figures of the drawings. Elements in the several figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures might be emphasized relative to other elements for facilitating understanding of the

various presently disclosed embodiments. In addition, common, but well-understood, elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present disclosure.

DETAILED DESCRIPTION

(22) In response to the problems described above, devices and methods are discussed herein that can implement semantic searching on SSD through a computational SSD system that distributes computing to each NAND flash die of the SSD, while the SSD controller handles the results aggregation with new on-die computation logic circuits to provide on device file semantic search. Specifically, many embodiments utilize computational SSD system that reads file feature vectors from multiple dies to the SSD controller, if there are millions of feature vectors to be compared, these feature vectors may be buffered in DRAM and controller handles the distance computing. Local, on-die AI/ML processing units may perform, for example, computation and comparison operations and pass the processing scores and results to the SSD controller. The SSD controller aggregates results from all dies and returns the result to the host. In some embodiments, the feature vector store size each on-die AI/ML processing unit may be configured as needed to adapt to different tasks and/or feature vector sizes. Based on this, the circuitry of AI/ML processing units and number of on-die AI/ML processing units may be configured and distributed as needed to provide increased computational speed or to meet certain die area constraints. Additional embodiments are discussed in more detail below.

(23) These solutions can help reduce substantial waste of bandwidth and energy from data movement between the host processor, the host memory, and the storage device. Moreover, these solutions can help reduce a bottleneck for the storage device controller in AI/ML processing where the storage device controller needs to handle both I/O and computing operations in AI/ML operations which requires reading the feature vectors from the multiple dies to the controller and buffering the feature vectors in memory to perform AI/ML computations and processing.

(24) Aspects of the present disclosure may be embodied as an apparatus, system, method, or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, or the like) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “function,” “module,” “apparatus,” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more non-transitory computer-readable storage media storing computer-readable and/or executable program code. Many of the functional units described in this specification have been labeled as functions, in order to emphasize their implementation independence more particularly. For example, a function may be implemented as a hardware circuit comprising custom VLSI circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A function may also be implemented in programmable hardware devices such as via field programmable gate arrays, programmable array logic, programmable logic devices, or the like.

(25) Functions may also be implemented at least partially in software for execution by various types of processors. An identified function of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified function need not be physically located together but may comprise disparate instructions stored in different locations which, when joined logically together, comprise the function and achieve the stated purpose for the function.

(26) Indeed, a function of executable code may include a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, across several storage devices, or the like. Where a function or portions of a function are implemented in software, the software portions may be stored on one or more computer-readable

and/or executable storage media. Any combination of one or more computer-readable storage media may be utilized. A computer-readable storage medium may include, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing, but would not include propagating signals. In the context of this document, a computer readable and/or executable storage medium may be any tangible and/or non-transitory medium that may contain or store a program for use by or in connection with an instruction execution system, apparatus, processor, or device.

(27) Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object-oriented programming language such as Python, Java, Smalltalk, C++, C #, Objective C, or the like, conventional procedural programming languages, such as the “C” programming language, scripting programming languages, and/or other similar programming languages. The program code may execute partly or entirely on one or more of a user's computer and/or on a remote computer or server over a data network or the like.

(28) A component, as used herein, comprises a tangible, physical, non-transitory device. For example, a component may be implemented as a hardware logic circuit comprising custom VLSI circuits, gate arrays, or other integrated circuits; off-the-shelf semiconductors such as logic chips, transistors, or other discrete devices; and/or other mechanical or electrical devices. A component may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices, or the like. A component may comprise one or more silicon integrated circuit devices (e.g., chips, die, die planes, packages) or other discrete electrical devices, in electrical communication with one or more other components through electrical lines of a printed circuit board (PCB) or the like. Each of the functions and/or modules described herein, in certain embodiments, may alternatively be embodied by or implemented as a component.

(29) A circuit, as used herein, comprises a set of one or more electrical and/or electronic components providing one or more pathways for electrical current. In certain embodiments, a circuit may include a return pathway for electrical current, so that the circuit is a closed loop. In another embodiment, however, a set of components that may not include a return pathway for electrical current may be referred to as a circuit (e.g., an open loop). For example, an integrated circuit may be referred to as a circuit regardless of whether the integrated circuit is coupled to ground (as a return pathway for electrical current) or not. In various embodiments, a circuit may include a portion of an integrated circuit, an integrated circuit, a set of integrated circuits, a set of non-integrated electrical and/or electrical components with or without integrated circuit devices, or the like. In one embodiment, a circuit may include custom VLSI circuits, gate arrays, logic circuits, or other integrated circuits; off-the-shelf semiconductors such as logic chips, transistors, or other discrete devices; and/or other mechanical or electrical devices. A circuit may also be implemented as a synthesized circuit in a programmable hardware device such as field programmable gate array, programmable array logic, programmable logic device, or the like (e.g., as firmware, a netlist, or the like). A circuit may comprise one or more silicon integrated circuit devices (e.g., chips, die, die planes, packages) or other discrete electrical devices, in electrical communication with one or more other components through electrical lines of a printed circuit board (PCB) or the like. Each of the functions and/or modules described herein, in certain embodiments, may be embodied by or implemented as a circuit.

(30) Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms

“including,” “comprising,” “having,” and variations thereof mean “including but not limited to”, unless expressly specified otherwise. An enumerated listing of items may not imply that any or all of the items are mutually exclusive and/or mutually inclusive, unless expressly specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise.

(31) Further, as used herein, reference to reading, writing, storing, buffering, and/or transferring data can include the entirety of the data, a portion of the data, a set of the data, and/or a subset of the data. Likewise, reference to reading, writing, storing, buffering, and/or transferring non-host data can include the entirety of the non-host data, a portion of the non-host data, a set of the non-host data, and/or a subset of the non-host data.

(32) Lastly, the terms “or” and “and/or” as used herein are to be interpreted as inclusive or meaning any one or any combination. Therefore, “A, B or C” or “A, B and/or C” mean “any of the following: A; B; C; A and B; A and C; B and C; A, B and C.” An exception to this definition will occur only when a combination of elements, functions, steps, or acts are in some way inherently mutually exclusive.

(33) Aspects of the present disclosure are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and computer program products according to embodiments of the disclosure. It will be understood that each block of the schematic flowchart diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a computer or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor or other programmable data processing apparatus, create means for implementing the functions and/or acts specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

(34) It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated figures. Although various arrow types and line types may be employed in the flowchart and/or block diagrams, they are understood not to limit the scope of the corresponding embodiments. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted embodiment.

(35) In the following detailed description, reference is made to the accompanying drawings, which form a part thereof. The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description. The description of elements in each figure may refer to elements of proceeding figures. Like numbers may refer to like elements in the figures, including alternate embodiments of like elements.

(36) Referring to FIG. 1, a schematic block diagram of an example host-computing device **110** with a storage system **102** suitable for computational SSD system supporting semantic search in accordance with an embodiment of the disclosure is shown. The computational SSD system **100** comprises one or more storage devices **120** of a storage system **102** within a host-computing device **110** in communication via a controller **126**. The host-computing device **110** may include a processor **111**, volatile memory **112**, and a communication interface **113**. The processor **111** may include one or more central processing units, one or more general-purpose processors, one or more application-specific processors, one or more virtual processors (e.g., the host-computing device **110** may be a virtual machine operating within a host), one or more processor cores, or the like. The communication interface **113** may include one or more network interfaces configured to

communicatively couple the host-computing device **110** and/or controller **126** of the storage device **120** to a communication network **115**, such as an Internet Protocol (IP) network, a Storage Area Network (SAN), wireless network, wired network, or the like.

(37) The storage device **120**, in various embodiments, may be disposed in one or more different locations relative to the host-computing device **110**. In one embodiment, the storage device **120** comprises one or more non-volatile memory devices **123**, such as semiconductor chips or packages or other integrated circuit devices disposed on one or more printed circuit boards, storage housings, and/or other mechanical and/or electrical support structures. For example, the storage device **120** may comprise one or more direct inline memory module (DIMM) cards, one or more expansion cards and/or daughter cards, a solid-state-drive (SSD) or other hard drive device, and/or may have another memory and/or storage form factor. The storage device **120** may be integrated with and/or mounted on a motherboard of the host-computing device **110**, installed in a port and/or slot of the host-computing device **110**, installed on a different host-computing device **110** and/or a dedicated storage appliance on the network **115**, in communication with the host-computing device **110** over an external bus (e.g., an external hard drive), or the like.

(38) The storage device **120**, in one embodiment, may be disposed on a memory bus of a processor **111** (e.g., on the same memory bus as the volatile memory **112**, on a different memory bus from the volatile memory **112**, in place of the volatile memory **112**, or the like). In a further embodiment, the storage device **120** may be disposed on a peripheral bus of the host-computing device **110**, such as a peripheral component interconnect express (PCI Express or PCIe) bus such, as but not limited to a NVM Express (NVMe) interface, a serial Advanced Technology Attachment (SATA) bus, a parallel Advanced Technology Attachment (PATA) bus, a small computer system interface (SCSI) bus, a FireWire bus, a Fibre Channel connection, a Universal Serial Bus (USB), a PCIe Advanced Switching (PCIe-AS) bus, or the like. In another embodiment, the storage device **120** may be disposed on a communication network **115**, such as an Ethernet network, an Infiniband network, SCSI RDMA over a network **115**, a storage area network (SAN), a local area network (LAN), a wide area network (WAN) such as the Internet, another wired and/or wireless network **115**, or the like.

(39) The host-computing device **110** may further comprise computer-readable storage medium **114**. The computer-readable storage medium **114** may comprise executable instructions configured to cause the host-computing device **110** (e.g., processor **111**) to perform steps of one or more of the methods disclosed herein. Additionally, or in the alternative, the buffering component **150** may be embodied as one or more computer-readable instructions stored on the computer-readable storage medium **114**.

(40) A device driver and/or the controller **126**, in certain embodiments, may present a logical address space **134** to the host clients **116**. As used herein, a logical address space **134** refers to a logical representation of memory resources. The logical address space **134** may comprise a plurality (e.g., range) of logical addresses. As used herein, a logical address refers to any identifier for referencing a memory resource (e.g., data), including, but not limited to: a logical block address (LBA), cylinder/head/sector (CHS) address, a file name, an object identifier, an inode, a Universally Unique Identifier (UUID), a Globally Unique Identifier (GUID), a hash code, a signature, an index entry, a range, an extent, or the like.

(41) A device driver for the storage device **120** may maintain metadata **135**, such as a logical to physical address mapping structure, to map logical addresses of the logical address space **134** to media storage locations on the storage device(s) **120**. A device driver may be configured to provide storage services to one or more host clients **116**. The host clients **116** may include local clients operating on the host-computing device **110** and/or remote clients **117** accessible via the network **115** and/or communication interface **113**. The host clients **116** may include, but are not limited to: operating systems, file systems, database applications, server applications, kernel-level processes, user-level processes, applications, and the like.

(42) In many embodiments, the host-computing device **110** can include a plurality of virtual machines which may be instantiated or otherwise created based on user-request. As will be understood by those skilled in the art, a host-computing device **110** may create a plurality of virtual machines configured as virtual hosts which is limited only on the available computing resources and/or demand. A hypervisor can be available to create, run, and otherwise manage the plurality of virtual machines. Each virtual machine may include a plurality of virtual host clients similar to host clients **116** that may utilize the storage system **102** to store and access data.

(43) The device driver may be further communicatively coupled to one or more storage systems **102** which may include different types and configurations of storage devices **120** including, but not limited to: solid-state storage devices, semiconductor storage devices, SAN storage resources, or the like. The one or more storage devices **120** may comprise one or more respective controllers **126** and non-volatile memory channels **122**. The device driver may provide access to the one or more storage devices **120** via any compatible protocols or interface **133** such as, but not limited to, SATA and PCIe. The metadata **135** may be used to manage and/or track data operations performed through the protocols or interfaces **133**. The logical address space **134** may comprise a plurality of logical addresses, each corresponding to respective media locations of the one or more storage devices **120**. The device driver may maintain metadata **135** comprising any-to-any mappings between logical addresses and media locations.

(44) A device driver may further comprise and/or be in communication with a storage device interface **139** configured to transfer data, commands, and/or queries to the one or more storage devices **120** over a bus **125**, which may include, but is not limited to: a memory bus of a processor **111**, a peripheral component interconnect express (PCI Express or PCIe) bus, a serial Advanced Technology Attachment (ATA) bus, a parallel ATA bus, a small computer system interface (SCSI), FireWire, Fibre Channel, a Universal Serial Bus (USB), a PCIe Advanced Switching (PCIe-AS) bus, a network **115**, Infiniband, SCSI RDMA, or the like. The storage device interface **139** may communicate with the one or more storage devices **120** using input-output control (IO-CTL) command(s), IO-CTL command extension(s), remote direct memory access, or the like.

(45) The communication interface **113** may comprise one or more network interfaces configured to communicatively couple the host-computing device **110** and/or the controller **126** to a network **115** and/or to one or more remote clients **117** (which can act as another host). The controller **126** may be part of and/or in communication with one or more storage devices **120**. Although FIG. **1** depicts a single storage device **120**, the disclosure is not limited in this regard and could be adapted to incorporate any number of storage devices **120**.

(46) The storage device **120** may comprise one or more non-volatile memory devices **123** of non-volatile memory channels **122**, which may include but is not limited to: ReRAM, Memristor memory, programmable metallization cell memory, phase-change memory (PCM, PCME, PRAM, PCRAM, ovonic unified memory, chalcogenide RAM, or C-RAM), NAND flash memory (e.g., 2D NAND flash memory, 3D NAND flash memory), NOR flash memory, nano random access memory (nano RAM or NRAM), nanocrystal wire-based memory, silicon-oxide based sub-10 nanometer process memory, graphene memory, Silicon Oxide-Nitride-Oxide-Silicon (SONOS), programmable metallization cell (PMC), conductive-bridging RAM (CBRAM), magneto-resistive RAM (MRAM), magnetic storage media (e.g., hard disk, tape), optical storage media, or the like. The one or more non-volatile memory devices **123** of the non-volatile memory channels **122**, in certain embodiments, comprise storage class memory (SCM) (e.g., write in place memory, or the like).

(47) While the non-volatile memory channels **122** may be referred to herein as “memory media,” in various embodiments, the non-volatile memory channels **122** may more generally comprise one or more non-volatile recording media capable of recording data, which may be referred to as a non-volatile memory medium, a non-volatile memory device, or the like. Further, the storage device **120**, in various embodiments, may comprise a non-volatile recording device, a non-volatile

memory array **129**, a plurality of interconnected storage devices in an array, or the like.

(48) The non-volatile memory channels **122** may comprise one or more non-volatile memory devices **123**, which may include, but may not limited to: chips, packages, planes, die, or the like. A controller **126** may be configured to manage data operations on the non-volatile memory channels **122**, and may comprise one or more processors, programmable processors (e.g., FPGAs), ASICs, micro-controllers, or the like. In some embodiments, the controller **126** may be configured to store data on and/or read data from the non-volatile memory channels **122**, to transfer data to/from the storage device **120**, and so on.

(49) The controller **126** may be communicatively coupled to the non-volatile memory channels **122** by way of a bus **127**. The bus **127** may comprise an I/O bus for communicating data to/from the non-volatile memory devices **123**. The bus **127** may further comprise a control bus for communicating addressing and other command and control information to the non-volatile memory devices **123**. In some embodiments, the bus **127** may communicatively couple the non-volatile memory devices **123** to the controller **126** in parallel. This parallel access may allow the non-volatile memory devices **123** to be managed as a group, forming a non-volatile memory array **129**. The non-volatile memory devices **123** may be partitioned into respective logical memory units (e.g., logical pages) and/or logical memory divisions (e.g., logical blocks). The logical memory units may be formed by logically combining physical memory units of each of the non-volatile memory devices **123**.

(50) The controller **126** may organize a block of word lines within a non-volatile memory device **123**, in certain embodiments, using addresses of the word lines, such that the word lines may be logically organized into a monotonically increasing sequence (e.g., decoding and/or translating addresses for word lines into a monotonically increasing sequence, or the like). In a further embodiment, word lines of a block within a non-volatile memory device **123** may be physically arranged in a monotonically increasing sequence of word line addresses, with consecutively addressed word lines also being physically adjacent (e.g., WL0, WL1, WL2, . . . WLN).

(51) The controller **126** may comprise and/or be in communication with a device driver executing on the host-computing device **110**. A device driver may provide storage services to the host clients **116** via one or more interfaces **133**. A device driver may further comprise a storage device interface **139** that is configured to transfer data, commands, and/or queries to the controller **126** over a bus **125**, as described above.

(52) Referring to FIG. 2, a schematic block diagram of an example storage device **120** suitable for computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. The controller **126** may include a front-end module **208** that interfaces with a host via a plurality of high priority and low priority communication channels, a back-end module **210** that interfaces with the non-volatile memory devices **123**, and various other modules that perform various functions of the storage device **120**. In some examples, each module may just be the portion of the memory that comprises instructions executable with the processor to implement the features of the corresponding module without the module including any other hardware. Because each module includes at least some hardware even when the included hardware comprises software, each module may be interchangeably referred to as a hardware module.

(53) The controller **126** may include a buffer management/bus control module **214** that manages buffers in random access memory (RAM) **216** and controls the internal bus arbitration for communication on an internal communications bus **217** of the controller **126**. A read only memory (ROM) **218** may store and/or access system boot code. Although illustrated in FIG. 2 as located separately from the controller **126**, in other embodiments one or both of the RAM **216** and the ROM **218** may be located within the controller **126**. In yet other embodiments, portions of RAM **216** and ROM **218** may be located both within the controller **126** and outside the controller **126**. Further, in some implementations, the controller **126**, the RAM **216**, and the ROM **218** may be located on separate semiconductor dies. As discussed below, in one implementation, the submission

queues and the completion queues may be stored in a controller memory buffer, which may be housed in RAM **216**.

(54) Additionally, the front-end module **208** may include a host interface **220** and a physical layer interface **222** that provides the electrical interface with the host or next level storage controller. The choice of the type of the host interface **220** can depend on the type of memory being used. Example types of the host interfaces **220** may include, but may not be limited to, SATA, SATA Express, SAS, Fibre Channel, USB, PCIe, and NVMe. The host interface **220** may typically facilitate transfer for data, control signals, and timing signals.

(55) The back-end module **210** may include an error correction engine **224** that encodes the data bytes received from the host and decodes and error corrects the data bytes read from the non-volatile memory devices **123** using error-correction code (ECC). The back-end module **210** may also include a command sequencer **226** that generates command sequences, such as program, read, and erase command sequences, to be transmitted to the non-volatile memory devices **123**.

Additionally, the back-end module **210** may include a RAID (Redundant Array of Independent Drives) module **228** that manages generation of RAID parity and recovery of failed data. The RAID parity may be used as an additional level of integrity protection for the data being written into the storage device **120**. In some cases, the RAID module **228** may be a part of the error correction engine **224**. A memory interface **230** provides the command sequences to the non-volatile memory devices **123** and receives status information from the non-volatile memory devices **123**. Along with the command sequences and status information, data to be programmed into and read from the non-volatile memory devices **123** may be communicated through the memory interface **230**. A flash control layer **232** may control the overall operation of back-end module **210**.

(56) Additional modules of the storage device **120** illustrated in FIG. 2 may include a media management layer **238**, which performs wear leveling of memory cells of the non-volatile memory devices **123**. The storage device **120** may also include other discrete components **240**, such as external electrical interfaces, external RAM, resistors, capacitors, or other components that may interface with controller **126**. In alternative embodiments, one or more of the RAID modules **228**, media management layer **238** and buffer management/bus control module **214** may be optional components that may not be necessary in the controller **126**.

(57) Finally, the controller **126** may also comprise an intelligent memory array logic **234**. In many embodiments, the intelligent memory array logic **234** can be configured to receive a query, extract contextual data from the received query, and determine a machine learning model for processing the query based on the extracted contextual data. The determination of a machine learning models for further processing the query can be done by controller **126**. In some embodiments, the controller **126** may determine one or more suitable machine learning models for further processing the query. The intelligent memory array logic **234** may be provided by hardware, firmware, software, electrical circuitry and/or components, or any combinations thereof for receiving data (e.g., a query) and performing machine learning processes, computations, comparisons, operations, or other processing to obtain and/or extract contextual data from the received query to generate one or more query vectors and then determine one or more suitable machine learning models for further processing the one or more query vectors. This can include accessing control data stored within the storage device **120**, scanning the non-volatile memory devices **123** for relevant pages and determining the relevant non-volatile memory devices **123** associated with the generated query vector, and passing the query vector to the one or more determined relevant non-volatile memory devices **123**.

(58) Each of the non-volatile memory devices **123** may include feature data and feature vector store relevant to one or more machine learning models and processes. The feature data may be distributed evenly within each of one or more non-volatile memory devices **123**. In some embodiments, non-volatile memory devices **123** may be grouped into a plurality of sets, where each set of non-volatile memory devices **123** provides a specific machine learning model and

performs machine learning processes based on the specific machine learning model. Moreover, each set of non-volatile memory devices **123** may comprise the same or different feature data and feature vector store. The controller **126** may then pass the query vector to one or more relevant non-volatile memory devices **123**.

(59) Each of the non-volatile memory devices **123** of memory array **129** may further include one or more machine learning processing units configured to process the query vector obtained from controller **126** as a first input. The feature data within one or more non-volatile memory device **123** may be utilized as a second input, and the query vector and feature data may be buffered in a memory device prior to processing by the one or more machine learning processing units. For example, processing the first and second inputs to generate a comparison value. The one or more non-volatile memory device **123** may be grouped into one or more memory sets based on similarity in machine learning processing units or similarity in feature data. The controller **126** may obtain either the comparison value or output data as a result of processing the first and second inputs from the one or more non-volatile memory device **123**. In some embodiments, one or more sets of non-volatile memory devices **123** may be grouped together based on particular machine learning model or processing, similarity in machine learning processing units, or similarity in feature data.

(60) Referring to FIG. 3, a conceptual illustration of a page of memory cells, organized for example in the NAND configuration, being sensed or programmed in parallel in accordance with an embodiment of the disclosure. FIG. 3 conceptually shows a bank of NAND strings **350** within a non-volatile memory device **123** of FIG. 1. A 'page' such as the page **360**, is a group of memory cells enabled to be sensed or programmed in parallel. This is accomplished in the peripheral circuits by a corresponding page of sense amplifiers **310**. The sensed results may be utilized in latches within a corresponding set of data latches **320**. Each sense amplifier can be coupled to a NAND string, such as NAND string **350** via a bit line **336**. For example, the page **360** may be along a row and is sensed by a sensing voltage applied to the control gates of the cells of the page connected in common to the word line WL3. Along each column, each memory cell such as memory cell **311** may be accessible by a sense amplifier via a bit line **336**. Data in the data latches **320** may be toggled in from or out to the memory controller **126** via a data I/O bus **331**.

(61) The NAND string **350** can be a series of memory cells, such as memory cell **311**, daisy-chained by their sources and drains to form a source terminal and a drain terminal respective at its two ends. A pair of select transistors S1, S2 can control the memory cell chain's connection to the external source via the NAND string's source terminal and drain terminal, respectively. In a memory array, when the source select transistor S1 is turned on, the source terminal is coupled to a source line **334**. Similarly, when the drain select transistor S2 is turned on, the drain terminal of the NAND string is coupled to a bit line **336** of the memory array. Each memory cell **311** in the chain acts to store a charge. It has a charge storage element to store a given amount of charge so as to represent an intended memory state. In many embodiments, a control gate within each memory cell can allow for control over read and write operations. Often, the control gates of corresponding memory cells of each row within a plurality of NAND strings may be all connected to the same word line (such as WL0, WL1 . . . WLn **342**). Similarly, a control gate of each of the select transistors S1, S2 (accessed via select lines **344** SGS and SGD respectively) provides control access to the NAND string via its source terminal and drain terminal, respectively.

(62) While the example memory device referred to above comprises physical page memory cells that store single bits of data, in most embodiments each cell is storing multi-bit data, and each physical page can have multiple data pages. Additionally, in further embodiments, physical pages may store one or more logical sectors of data. Typically, the host-computing device **110** (see FIG. 1) operating with a disk operating system manages the storage of a file by organizing the content of the file in units of logical sectors, which is typically in one or more units of 512 bytes. In some embodiments, a physical page may have 16 kB of memory cells being sensed in parallel by corresponding 16 kB of sense amplifiers via 16 kB of bit lines. An example logical sector assigned

by the host has a size of 2 kB of data. Thus, a physical page can store 8 sectors if the cells are each configured to store 1 bit of data (SLC). For MLC, TLC, and QLC and other increased density structures, each cell can store 2, 3, 4 or more bits of data, and each physical page can store 16, 32, 64 or more logical sectors depending on the structure utilized. For feature vector size and allocation in the computational SSD system supporting semantic search, the machine learning processing unit of the non-volatile memory device **123** may be configured to be able to adapt to different feature vector size. As an example, a media file (e.g., image, document, video, etc.,) may be represented by any number of feature vectors. In some embodiments, a feature vector for a media file could be 128-1024 features, the size of a feature vector may be 256B-2 KB. Then for a 16 KB NAND flash page, reading a page will get 8-64 feature vectors.

(63) One unique difference between flash memory and other types of memory is that a memory cell must be programmed from an erased state which is associated with no charge within the memory cell. This requires that the floating gate must first be emptied of charge prior to programming. Programming adds a desired amount of charge back to the floating gate. It may not support removing a portion of the charge from the floating to go from a more programmed state to a lesser one. Thus, new data cannot overwrite existing data and must be written to a previously unwritten or erased location. Furthermore, erasing all the charges from a floating gate can often take an appreciable amount of time. For that reason, it will be cumbersome and inefficient to erase cell by cell or even page by page. Therefore, in most embodiments, the array of memory cells is often divided into a large number of blocks. As is common in many flash-based memory systems, the block is the unit of erase. That is, each block contains the minimum number of memory cells that may be erased in one action. This combined with the limited lifespans of memory cells within the flash memory increases the desire to limit the amount of erasing and programming occurring within the storage device.

(64) Referring to FIG. 4, a schematic block diagram of an example computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. As discussed above, controller **426** includes intelligent memory array logic **434** and results aggregation unit **412**. The controller **426** receives a semantic search query **401** and passes the query **401** to intelligent memory array logic **434** for processing. In some embodiments, the controller **426** may receive data (e.g., a query) and perform machine learning processes, computations, comparisons, operations, or other processing to obtain and/or extract contextual data from the received query **401**, and prepare the query **401** for intelligent memory array logic **434**. The contents of query **401** may include one or a plurality of text, documents, images, audio, or other media. Upon receiving query **401**, controller **426** processes the query **401** and extracts contextual data from the query **401**. In addition to extracting contextual data from query **401**, other relevant contextual information for building contextual data may be obtained by comparison of the query **401** and/or contextual data with one or more dictionaries, libraries, or databases, or a combination thereof. The relevant contextual information may be added to query **401** or the extracted contextual data from query **401** for further processing. Moreover, the contextual information may be stored locally on controller **426** or remotely accessible to the controller **426** and intelligent memory array logic **434**. The controller **426** passes the contextual data to intelligent memory array logic **434**, and the intelligent memory array logic **434** makes a determination of the machine learning model for processing query **401** based on the contextual data provided by controller **426**.

(65) In some embodiments, the controller **426** may determine one or more suitable machine learning models or machine learning processes within intelligent memory array logic **434** for further processing the query **401**. The intelligent memory array logic **434** may receive data (e.g., a query) and perform machine learning processes, computations, comparisons, operations, or other processing to obtain and/or extract contextual data from the received query **401**.

(66) The intelligent memory array logic **434** processes the query **401** and/or contextual data from query **401** and generates a query vector **403** based on the received query **401**, the context data from

the query **401**, or a combination thereof. Further, the query vector **403** may be generated, in part, or in whole, based on the determined machine learning model. The intelligent memory array logic **434** may determine one or more relevant non-volatile memory devices **423** or sets of non-volatile memory devices **423** for processing the generated query vector **403**. The determination may be based on the query **401**, contextual data from query **401**, relevant contextual information from the query **401**, or the determined machine learning model, or any combination thereof. The intelligent memory array logic **434** then passes the generated query vector **403** to one or more relevant non-volatile memory devices **423**.

(67) As discussed above, each of the one or more non-volatile memory devices **423** comprise feature data and one or more machine learning processing units **421aa**, **421ab**, . . . **421an**, **421ba**, **421bb**, . . . **421bn**, **421na**, **421nb**, . . . **421nn** . . . etc., (hereinafter, “machine learning processing unit **421**”) where each machine learning processing unit **421** may include hardware, firmware, software, electrical circuitry and/or components, or any combinations thereof for performing machine learning processes, computations, comparisons, operations, or other processing to obtain a result and/or value based on input(s) from feature data stored on non-volatile memory devices **423** and input(s) provided by the intelligent memory array logic **434**. Each of the one or more machine learning processing units **421** may be configured to process the passed in query vector **403** as a first input and utilize stored feature data as a second input. The one or more machine learning processing units **421** may then process the first and second inputs to generate a result and/or value (e.g., a comparison value) as an output **405** to controller **426**. The intelligent memory array logic **434** may add feature metadata to output **405** for further processing by the controller **426**.

(68) In some embodiments, the query vector **403** and feature data may be buffered in a memory device prior to processing by the one or more machine learning processing units **421**. Moreover, the feature data may be distributed evenly within the one or more non-volatile memory devices **423**. In some embodiments, each of the one or more non-volatile memory devices **423** may be grouped into one or more memory sets based on similarity in machine learning processing units or similarity in feature data.

(69) The results aggregation unit **412** obtains results and/or value from all machine learning processing unit **421** as output **405** and performs machine learning processes, computations, comparisons, operations, or other processing on output **405**, such as for example, searching, gathering, and presenting results from output **405** in a summarized, report-based, or tabular form as result **470**. The result **470** may be passed to the host **110** for further processing.

(70) Referring to FIG. 5, a schematic block diagram of an example machine learning processing unit of the computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. In some embodiments, the machine learning processing unit **421** may be configured to trade off area for speed, where speed is limited, yielding less hardware reuse or parallel processing with lower area efficiency. As an example, the machine learning processing unit **500** may be configured for higher speed. The machine learning processing unit **500** may include hardware, firmware, software, electrical circuitry and/or components, or any combinations thereof for performing machine learning processes, computations, comparisons, operations, or other processing. The machine learning processing unit **500** includes module **571** that determines the feature vector size of query vector $V_{sub.q}$, one or more distance compute units **573a** . . . **573n** (hereinafter, “distance compute unit **573**”) that compute the distance (e.g., Euclidean/Hamming distance) between two input vectors (e.g., $V_{sub.q}$, $V_{sub.f(i)}$), control unit **574** that receives the feature size extracted from query vector V by module **571** and adapts to the variable feature vector size by, for example, comparing the number of distance comparisons to the feature vector size and choosing the correct one as the final similarity score of all the input feature vectors $V_{sub.f(i)}$. . . $V_{sub.f(i+n)}$. Further, the machine learning processing unit **500** includes module **575** having one or more machine learning operators, functions, or registers to perform machine learning processing, for example, comparing two distance inputs from the distance compute units **573** and outputting the

smaller distance to a register. The register of module **575** may then provide the result, score, or value to non-volatile memory devices **423** and/or controller **426**.

(71) Referring to FIG. **6**, a schematic block diagram of an example machine learning processing unit of the computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. In some embodiments, the machine learning processing unit **421** may be configured to trade off speed for area, where area is limited, yielding hardware reuse or serial processing with lower speed. As an example, the machine learning processing unit **600** may be configured for high area efficiency. The machine learning processing unit **600** may include hardware, firmware, software, electrical circuitry and/or components, or any combinations thereof for performing machine learning processes, computations, comparisons, operations, or other processing. The machine learning processing unit **600** includes module **671** that determines the feature vector size of query vector $V_{sub.q}$, distance compute unit **673** for computing the distance (e.g., Euclidean/Hamming distance) between two input vectors (e.g., $V_{sub.q}$, $V_{sub.f}$), control unit **674** that receives the feature size extracted from query vector $V_{sub.q}$ by module **671** and adapts to the variable feature vector size by, for example, comparing the number of distance comparisons to the feature vector size and choosing the correct one as the final similarity score of all the input feature vectors $V_{sub.f}$. Further, the machine learning processing unit **600** includes module **675** having one or more machine learning operators, functions, or registers to perform machine learning processing, for example, comparing two distance inputs from the distance compute units **673** and outputting the smaller distance to a register. The register of module **675** may then provide the result, score, or value to non-volatile memory devices **423** and/or controller **426**.

(72) Referring to FIG. **7**, a flowchart depicting a process **700** for utilizing an example computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. The process **700** can begin by receiving a query (block **710**). In many embodiments, this occurs the host or SSD controller receives a query for a semantic search. In some embodiments, the user provides a semantic search request that is passed to the host or SSD controller. In response, a variety of embodiments of the controller may direct a plurality of processes, such as those within the controller **426** for computational SSD system supporting semantic search (see FIG. **4**) to execute the semantic search process **700**.

(73) The process **700** can extract contextual data from the received query (block **715**). In some embodiments, the contextual data may be obtained directly from the query. As described above, in some embodiments, the contextual information relevant to the query may be obtained through other resources to help build contextual data for the query. Relevant contextual information for building contextual data may be obtained by comparison of the query and/or contextual data with one or more machine learning processes having access to dictionaries, libraries, or databases, or a combination thereof.

(74) The process **700** can determine a machine learning model based on the extracted contextual data (block **720**). As known in the art, various machine learning models may be used to extract and process features from the query, for example, convolutional neural network (CNN), computer vision, supervised learning, semi-supervised learning, unsupervised learning, reinforcement learning (RL), deep learning (DL), Artificial neural networks (ANNs), computer vision, speech recognition, natural language processing, machine translation, and others as desired.

(75) Upon determination of a machine learning model based on the extracted contextual data, the process **700** may utilize the machine learning model to process the query (block **725**). By way of example and not limitation, in processing the query, various features may be obtained from the machine learning model, the features may be processed to create a vector and prepared for machine learning processes such as a similarity search.

(76) Based on the features obtained from the received query and the determined machine learning model, the process **700** can generate a query vector (block **730**) and determine one or more relevant non-volatile memory devices associated with the generated query vector (block **735**). As discussed

above, the non-volatile memory device **123** may be part of any suitable storage device **120** that may have a variety of different types of memory devices. By way of example, and not limitation, a storage device may comprise a plurality of non-volatile NAND memory devices including single-level cell NAND memory and quad-level cell NAND memory.

(77) The process **700** can pass in the query vector to the one or more determined relevant non-volatile memory devices (block **740**). The process **700** can process the passed in query vector as a first input (block **745**) and utilize the feature data on the one or more relevant non-volatile memory devices as a second input (block **750**). As described above, the one or more determined relevant non-volatile memory devices have machine learning processing units for processing the query vector as a first input and the feature data stored on one or more non-volatile memory devices as the second input.

(78) The process **700** can process the first and second inputs to generate a comparison value (block **755**). In certain embodiments, the process **700** may utilize hyperdimensional vectors/vector symbolic architecture to enable on-die calculation without error-correction code (ECC). In this way, features can be stored as hyperdimensional vectors rather than dense feature vectors which may be noise resilient.

(79) Referring to FIG. **8**, a flowchart depicting a process **800** for utilizing an example computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. The process **800** can begin by processing a query using a machine learning model (block **810**). In many embodiments, this occurs the host or SSD controller receives a query for a semantic search. In some embodiments, the user provides a semantic search request that may be passed to the host or SSD controller. In response, a variety of embodiments of the controller may direct a plurality of processes, such as those within the controller **426** for computational SSD system supporting semantic search (see FIG. **4**) to execute the semantic search process **800**.

(80) The process **800** can generate a query vector (block **815**). In generating a query vector, contextual data may be obtained directly from the query and processing of the query by the machine learning model. By way of example and not limitation, in processing the query, various features may be obtained from the machine learning model, the features may be processed to create a vector and prepared for machine learning processes such as a similarity search. As known in the art, various machine learning models may be used to extract and process features from the query, for example, convolutional neural network (CNN), computer vision, supervised learning, semi-supervised learning, unsupervised learning, reinforcement learning (RL), deep learning (DL), Artificial neural networks (ANNs), computer vision, speech recognition, natural language processing, machine translation, and others as desired.

(81) The process **800** can pass in the query vector to one or more machine learning processing units as a first unit (block **820**) that further utilizes feature data stored on one or more relevant non-volatile memory devices as a second input (block **825**). The process **800** can utilize the one or more machine learning processing units to process the first and second inputs to determine an output data to retrieve from the one or more non-volatile memory devices (block **830**). Process **800** can assign feature metadata to the output data using the intelligent memory array logic (block **835**). In certain embodiments, the process **800** may utilize hyperdimensional vectors/vector symbolic architecture to enable on-die calculation without error-correction code (ECC). In this way, features can be stored as hyperdimensional vectors rather than dense feature vectors which may be noise resilient.

(82) Referring to FIGS. **7-8**, hyperdimensional computing for calculating similarity among data and may be realized by, for example, three operations, including addition, multiplication, and permutation. Hyperdimensional computing may be inherently robust as information is evenly distributed over every bit of the hypervectors and provides fast learning ability, high energy efficiency and acceptable accuracy in learning and classification tasks.

(83) Referring to FIG. **9**, a schematic block diagram of an example machine learning process in accordance with an embodiment of the disclosure. As shown an input for a machine learning

system **900** may include semantic search request by one or more queries **901**. The query **901** may include text, image, document, audio, or other media. The query **901** may be passed to an intelligent memory array logic **934** that performs machine learning processes, computations, comparisons, operations, or other processing to obtain and/or extract contextual data from the received query **901**. The intelligent memory array logic **934** determines contextual data from query **901** and assigns features to the query **901** to forming query vector **903** as a first input for a machine learning processing unit (MPLU) **921** of machine learning system **900**.

(84) A second input for the machine learning system **900** includes stored data on SSD device such as one or more media files **902**. The media file **902** may include text, image, document, audio, or other media. The stored media files **902** may be assigned feature data **904** by intelligent memory array logic **934** and each media file **902** can be stored on one or more non-volatile memory devices **123**. The feature data **904** may be scanned by the MPLU **921** from one or more non-volatile memory devices **123** as the second input for comparison with the query vector **903**.

(85) The MPLU **921** follows an iterative process of retrieving, extracting, calculating, and comparing feature data **904** of stored media files **902** with query vector **903** to obtain output data **905**. The output data **905** can be provided as a result of one or more operations by the MPLU **921**. In some embodiments, the output data **905** may be a similarity score of the comparison between the query vector **903** and one or more sets of feature data **904**. The output data **905** may then be passed to the controller **126** for further processing.

(86) In some embodiments, the controller **426** may determine one or more suitable machine learning models or machine learning processes within intelligent memory array logic **934** for further processing the query **401**. The intelligent memory array logic **934** may receive data (e.g., a query) and perform machine learning processes, computations, comparisons, operations, or other processing to obtain and/or extract contextual data from the received query **901**.

(87) As discussed above, each machine learning processing unit **921** may include hardware, firmware, software, electrical circuitry and/or components, or any combinations thereof for performing machine learning processes, computations, comparisons, operations, or other processing to obtain a result and/or value based on input(s) from feature data stored on non-volatile memory devices **123** and input(s) provided by the intelligent memory array logic **934**.

(88) In some embodiments, the query vector **903** and feature data **904** may be buffered in a memory device prior to processing by the one or more machine learning processing units **921**. Moreover, the feature data **904** may be distributed evenly within the one or more non-volatile memory devices **123**. In some embodiments, each of the one or more non-volatile memory devices **123** may be grouped into one or more memory sets based on similarity in machine learning processing units or similarity in feature data.

(89) Referring to FIG. **10A**, a schematic block diagram of an example error correction system in accordance with an embodiment of the disclosure. As shown an error correction system **1000** for integrating ECC functions for encoding and decoding dense feature vectors.

(90) The example error correction system **1000** may include host **1010**, SSD controller **1026**, NAND flash memory array **1029**, and volatile memory **1016** (e.g., DRAM). The error correction system **1000** may further include host interface **1020**, embedded processor **1011**, ECC memory module **1024** having at least one encoding **1024a** and decoding **1024b** circuitry and processing, and flash interface **1032** included, in part or in whole, or otherwise in communication with or integrated within SSD controller **1026**. The embedded processor **1011** may be in communication with volatile memory **1016**. The volatile memory **1016** may be integrated within SSD controller **1026** and/or in communication with embedded processor **1011**.

(91) The NAND flash memory array **1029** may include a plurality of NAND dies **1023**, each NAND die having memory circuitry and logic **1027** that includes ECC memory module **1028** and machine learning processing unit **1021**. In some embodiments, the example error correction system **1000** may operate in storage mode where SSD controller **1026** receives file data and one or more

instructions from host **1010** to store data on NAND dies **1023**. Further, in storage mode, SSD controller **1026** may receive one or more instructions from host **1010** to access the same or different file data stored on NAND dies **1023**. As shown, in write operation, ECC memory module **1024** may encode error correction code within file data to facilitate storage, retrieval and processing of error free file data on NAND dies **1023**. Further, to retrieve error free file data to send to host **1010**, in read operation, ECC memory module **1024** may process and/or remove file data errors received from file data stored on NAND dies **1023** using the encoded error correction code for the file data. In some embodiments, file data errors in storage mode may be introduced, for example and not by limitation, at flash interface **1032** or at any file data processing stage within SSD controller **1026** from receiving file data and/or instructions by host **1010** to storing file data and/or instructions on NAND flash memory array **1029**.

(92) In some embodiments, the example error correction system **1000** may operate in compute mode where NAND flash memory array **1029** utilizes memory circuitry and logic **1027** having ECC memory module **1028** to encode, decode, verify, and provide error free feature data (e.g., dense feature vector) to machine learning processing unit **1021** to compare with a query vector to obtain a score. Thus, each NAND die **1023** includes ECC functions for encoding and decoding feature data error correction code to provide error free feature data or error free dense feature vectors. Thus, NAND dies **1023** store at least two types of data: raw file data and feature data (e.g., feature vectors) corresponding to the file. The memory circuitry and logic **1027** may perform feature data management to maximize feature data processing efficiency, for example, parallel processing of feature data and on-die. In some embodiments, similar to machine learning processing units **1021** being disposed in each NAND die **1023**, memory circuitry and logic **1027** may also be distributed in NAND flash memory array **1029** as single units per each NAND die **1023** to handle file data and feature data stored on the corresponding NAND die **1023**. Moreover, in some embodiments, feature data may be equally partitioned and distributed across each NAND die **1023** of NAND flash memory array **1029**. Further, the feature vectors may be relocated within the same NAND die **1023** during storage management operations such as, for example, garbage collection.

(93) FIG. **10B** is a chart of the example error correction system of FIG. **10A** in accordance with an embodiment of the disclosure. To ensure data integrity, data scrubbing may be performed on feature vectors (e.g., using FV scrubbing) and file data (e.g., using conventional data scrubbing). As shown, the number of bit flips per time for FV data scrubbing on feature data may be performed at higher frequency than conventional data scrubbing on file data or at a fraction of the frequency of conventional data scrubbing, for example, $\frac{1}{6}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$ conventional scrubbing time. Moreover, FV scrubbing and conventional scrubbing may be triggered once a predetermined threshold of bit flips is reached. For example, when an FV threshold is reached FV scrubbing can be performed and when a conventional threshold is reached conventional scrubbing may be performed. In some embodiments, a combination of time/frequency and threshold may be used to perform at least one of FV scrubbing or conventional scrubbing. As shown, FV data scrubbing may be performed at $\frac{2}{3}$ the time of a convention scrubbing of file data.

(94) FIG. **10C** is a table of the example error correction system of FIG. **10A** in accordance with an embodiment of the disclosure. File data and feature data error correction is shown as described above. File data and feature data may also undergo data scrubbing to reduce the likelihood of single correctable errors accumulating into numerous uncorrectable errors. As shown, an error correction engine can be used to verify file data and conventional data scrubbing may be used for wear leveling to maintain data integrity of file data in storage mode. Conventional data scrubbing may include error correction techniques that use one or more background tasks to periodically inspect main memory or storage for errors, then correct detected errors using redundant data in the form of different checksums or copies of data. The file data is cleansed through a process of detecting and correcting corrupt or inaccurate records from a record set, table, or database and identifying

incomplete, incorrect, inaccurate, or irrelevant parts of the data and then replacing, modifying, or deleting the erroneous, corrupt, or coarse data.

(95) As an example, when the example error correction system **1000** is in compute mode for feature vector data (e.g., dense feature vectors) on-die ECC is required. In some embodiments, to reduce the area and latency overhead of on-die ECC, weaker ECC like Bose-Chaudhuri-Hocquenghem (BCH) may be used. In some embodiments, other types of on-die ECC may be used, for example, Low Density Parity Check (LDPC). Moreover, on-die ECC may further be used for file data error correction as well. In some embodiments, feature vector data (e.g., dense feature vectors) may be stored in their own memory blocks separate from file data to facilitate easier data management. In some embodiments, feature vector data (e.g., dense feature vectors) may be stored with relevant file data in the same memory block to facilitate faster compute times and data access. Data scrubbing of file data may use conventional data scrubbing described above, while feature data may use a separate data scrubbing algorithm to, for example, cleanup erroneous, incorrect, incomplete, empty, or duplicate feature vector data.

(96) In some embodiments, when SSD storage device **120** is operating in semantic search mode, requests from host **1010** to serve read/write requests may be ignored, delayed, and performed after some semantic search operations have been completed, or queued to be performed after all semantic search operations have been completed, or any combination thereof. Thus, volatile memory **1016** may be used to load feature data related to data management data structures, rather than logical-to-physical (L2P) table for file data accesses.

(97) In regard to feature vector data management, an application needs to maintain at least one database in SSD storage device **120** for performing fast semantic search, for example, the database may comprise of attributes: filename, hashvalue, and feature vector data. Moreover, at least one file may contain the attributes filename and hashvalue that will be loaded into the application when launching the application. In some embodiments, the information of hashvalue and feature vector data may be contained in one or more data structures (e.g., Table X) to index one or more feature vectors. In some embodiments, Table X may be stored in the SSD storage device **120** firmware, and Table X may comprise of attributes of Addr, hashvalue, and feature vector data. Further, according to the returned hashvalue(s) from the semantic search, the application may retrieve filename(s) from the search result. In some embodiments, Table X size may be proportional to the number of files stored in SSD storage device **120**. Further, in some embodiments, Addr may be in the form of [Die, Plane, Block, Page, Segment] where the number of bit to represent Segment.

(98)
$$\text{Segment} = \log_2 \left(\frac{\text{pagesize}}{\text{featurevectorsize} + \text{hashvaluesize}} \right).$$

(99) In some embodiments, the hashvalue feature vector may be the concatenation of the hashvalue and its feature vector to simplify data management. Moreover, when SSD storage device **120** receives the search request (e.g., a query), the SSD controller **1026** needs to generate the special search command(s) and send them each NAND flash die **1023**. The SSD controller **1026** may inform the NAND flash die **1023** which block(s) contains the feature vector data and the NAND flash die **1023** should perform the computation accordingly. Moreover, each NAND flash die **1023** may perform the function of top_k hashvalue=Loop(search(query_vec, plane #, block #, last_page #, last segment #)).

(100) FIG. **11A** is a schematic block diagram of an example error correction system in accordance with an embodiment of the disclosure. As shown an error correction system **1100** for integrating ECC functions for encoding and decoding Hyper Dimensional Vectors (HDV) or Vector Symbolic Architecture (VSA) where on-die ECC may be optional. In the present disclosure HDV and VSA may be used interchangeably, however, hereinafter HDV will be used. In some embodiments, integrating ECC logic, circuitry, or processing in each NAND die **1123** may be too costly. For example, incorporating an ECC memory module **1128** in each NAND die **1123** may be prohibitively expensive, or take up too much area on the NAND flash memory array **1129**. Instead, HDV/VSA may be stored for each file, where such contextual/semantic representations may be

robust to noise, preserving similarities, and may be read from NAND flash dies **1123** and computed without error corrections.

(101) The example error correction system **1100** may include host **1110**, SSD controller **1126**, NAND flash memory array **1129**, and volatile memory **1116** (e.g., DRAM). The error correction system **1100** may further include host interface **1120**, embedded processor **1111**, ECC memory module **1124** having at least one encoding **1124a** and decoding **1124b** circuitry and processing, and flash interface **1132** included, in part or in whole, or otherwise in communication with or integrated within SSD controller **1126**. The embedded processor **1111** may be in communication with volatile memory **1116**. The volatile memory **1116** may be integrated within SSD controller **1126** and/or in communication with embedded processor **1111**.

(102) The NAND flash memory array **1129** may include a plurality of NAND dies **1123**, each NAND die having memory circuitry and logic **1127** that includes machine learning processing unit **1121**. In some embodiments, the example error correction system **1100** may operate in storage mode where SSD controller **1126** receives file data and one or more instructions from host **1110** to store data on NAND dies **1123**. Further, in storage mode, SSD controller **1126** may receive one or more instructions from host **1110** to access the same or different file data stored on NAND dies **1123**. As shown, in write operation, ECC memory module **1124** may encode error correction code within file data to facilitate storage, retrieval and processing of error free file data on NAND dies **1123**. Further, to retrieve error free file data to send to host **1110**, in read operation, ECC memory module **1124** may process and/or remove file data errors received from file data stored on NAND dies **1123** using the encoded error correction code for the file data. In some embodiments, file data errors in storage mode may be introduced, for example and not by limitation, at flash interface **1132** or at any file data processing stage within SSD controller **1126** from receiving file data and/or instructions by host **1110** to storing file data and/or instructions on NAND flash memory array **1129**.

(103) In some embodiments, the example error correction system **1100** may operate in compute mode where NAND flash memory array **1129** utilizes memory circuitry and logic **1127** and ECC memory module **1128** of SSD controller **1126** to encode, decode, verify, and provide error free feature data (e.g., HDV/VSA vector) to machine learning processing unit **1121** to compare with a query vector to obtain a score. Each NAND die **1123** utilizes two copies of HDV vectors, ECC functions from ECC memory module **1128**, and an example error correction scheme described below to encode and decode feature data error correction code to provide error free feature data, for example, error free HDV vectors.

(104) In the example error correction system **1100**, NAND dies **1123** store at least two types of data: raw file data and HDV/VSA vectors. The feature data (e.g., HDV vectors) corresponding to the file data may be very robust to noise, preserve similarities, and could be read from NAND dies **1123** and computed without error correction. In some embodiments, two copies of HDV vectors can be stored on each NAND die **1123** corresponding to the file data; a first copy, HDV without ECC coding and a second copy, HDV with SSD controller **1026** ECC encoding. As shown, the first copy of HDV vectors without ECC coding may be used by machine learning processing unit **1121**. However, since there is up limitation on its resilience, blocks storing HDV vectors need to be wear leveled as well.

(105) The second copy of HDV vectors with ECC coding can be used for correcting the errors in the first copy of HDV vectors. When a scrubbing algorithm finds the bit error rate of the HDV copy with ECC coding over the predefined threshold, corrected HDV without ECC coding may be stored on memory circuitry and logic **1127**, and the HDV copy with ECC coding can be relocated onto on memory circuitry and logic **1127**. In some embodiments, depending on the application, when VSA vectors are very long and super robust, a separate scrubbing algorithm for the second copy of HDV vectors with ECC coding may be not needed, otherwise, a separate scrubbing algorithm can be performed on HDV vectors with ECC coding.

(106) The memory circuitry and logic **1127** may perform feature data management to maximize feature data processing efficiency, for example, parallel processing of feature data and on-die. In some embodiments, memory circuitry and logic **1127** may also be distributed in NAND flash memory array **1129** as single units per each NAND die **1123** to handle file data and HDV vector data stored on the corresponding NAND die **1123**. Moreover, in some embodiments, feature data may be equally partitioned and distributed across each NAND die **1123** of NAND flash memory array **1129**. Further, the feature vector data may be relocated within the same NAND die **1123** during storage management operations such as, for example, garbage collection.

(107) FIG. **11B** is a chart of the example error correction system of FIG. **11A** in accordance with an embodiment of the disclosure. To ensure data integrity, data scrubbing may be performed on HDV vectors (e.g., using HDV scrubbing) and file data (e.g., using conventional data scrubbing). As shown, the number of bit flips per time for HDV data scrubbing on feature data may be performed at higher frequency than conventional data scrubbing on file data or at a fraction of the frequency of conventional data scrubbing, for example, $\frac{1}{6}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$ conventional scrubbing time. Moreover, HDV scrubbing and conventional scrubbing may be triggered once a predetermined threshold of bit flips is reached. For example, when an HDV threshold is reached HDV scrubbing can be performed and when a conventional threshold is reached conventional scrubbing may be performed. In some embodiments, a combination of time/frequency and threshold may be used to perform at least one of HDV scrubbing or conventional scrubbing. As shown, HDV data scrubbing may be performed at $\frac{2}{3}$ the time of a convention scrubbing of file data.

(108) FIG. **11C** is a table of the example error correction system of FIG. **11A** in accordance with an embodiment of the disclosure. File data and feature data error correction is shown as described above. File data and feature data may also undergo data scrubbing to reduce the likelihood of single correctable errors accumulating into numerous uncorrectable errors. As shown, SSD controller **1126** ECC may be used to verify file data and conventional data scrubbing may be used, for example, Low Density Parity Check (LDPC) for wear leveling to maintain data integrity of file data in storage mode. Conventional data scrubbing may include error correction techniques use one or more background tasks to periodically inspect main memory or storage for errors, then correct detected errors using redundant data in the form of different checksums or copies of data. The file data can be cleansed through a process of detecting and correcting corrupt or inaccurate records from a record set, table, or database and identifying incomplete, incorrect, inaccurate, or irrelevant parts of the data and then replacing, modifying, or deleting the erroneous, corrupt, or coarse data.

(109) As an example, when the example error correction system **1100** is in compute mode, conventional data scrubbing may be not needed for the first copy of HDV vectors, (i.e., HDV, HDV vectors without ECC coding). In some embodiments, conventional data scrubbing may be used as necessary for the first copy of HDV vectors to maintain data integrity of feature data for the first copy of HDV vectors without ECC coding. Further, in compute mode, SSD controller **1126** ECC can be used to verify feature data of the second copy of HDV vectors, (i.e., HDV-COPY, HDV vectors with ECC coding) using, for example, Low Density Parity Check (LDPC). Moreover, at least one of conventional data scrubbing or HDV scrubbing using feature data scrubbing algorithm may be used to, for example, cleanup erroneous, incorrect, incomplete, empty, or duplicate feature vector data.

(110) In some embodiments, when SSD storage device **120** is operating in semantic search mode, requests from host **1110** to serve read/write requests may be ignored, delayed, and performed after some semantic search operations have been completed, or queued to be performed after all semantic search operations have been completed, or any combination thereof. Thus, volatile memory **1116** may be used to load feature data related to data management data structures, rather than logical-to-physical (L2P) table for file data accesses. Moreover, HDV vector data management may be performed as described above.

(111) In some embodiments, the hashvalue feature vector may be the concatenation of the

hashvalue and its feature vector to simplify data management. Moreover, when SSD storage device **120** receives the search request (e.g., a query), the SSD controller **1126** needs to generate the special search command(s) and send them each NAND flash die **1123**. The SSD controller **1126** may inform the NAND flash die **1123** which block(s) contains the feature vector data and the NAND flash die **1123** should perform the computation accordingly. Moreover, each NAND flash die **1123** may perform the function of top_k hashvalue=Loop(search(query_vec, plane #, block #, last_page #, last segment #)).

(112) FIG. **12** is a schematic block diagram of an example error correction mode and storage mode of the computational SSD system **1200** supporting semantic search with error correction systems in accordance with an embodiment of the disclosure. As shown the example computational SSD system **1200** supports semantic search in compute mode and raw file data processing in storage mode with host **1210** providing respective instructions and file data or feature data to SSD storage device **1220**. In some embodiments, where non-volatile memory **1316** is limited, SSD storage device **1320** could switch between compute mode and storage mode to achieve the result file retrieval. Moreover, requests from host **1310** to serve read/write requests may be ignored, delayed, and performed after some semantic search operations have been completed, or queued to be performed after all semantic search operations have been completed, or any combination thereof. Thus, volatile memory **1116** may be used to load feature data related to data management data structures, rather than logical-to-physical (L2P) table for file data accesses.

(113) In step **1**: compute mode, non-volatile memory **1216** (e.g., DRAM) may be used by Table X, the query **1201** may be sent to SSD storage device **1220**, processed (e.g., machine learning processing unit **1021/1121**) and then result or hash **1203** returned to the host **1210** for the semantic search request.

(114) In step **2**: storage mode, non-volatile memory **1216** (e.g., DRAM) may be used for L2P table mapping, the data request **1205** can be sent to SSD storage device **1220**, processed (e.g., by SSD controller **1026/1126**) and then file data **1207** returned to the host **1210** for the file search/read/write/access request.

(115) FIG. **13** is a schematic block diagram of another example error correction mode and storage mode of the computational SSD system **1300** supporting semantic search with error correction systems in accordance with an embodiment of the disclosure. In some embodiments, where non-volatile memory **1316** may be sufficient for both storage and compute modes, storage and compute operations may be done concurrently or in parallel without requiring separate modes.

(116) For example, the non-volatile memory **1316** (e.g., DRAM) may be used by Table X, the query **1301** can be sent to SSD storage device **1320**, processed (e.g., machine learning processing unit **1021/1121**) and then result or hash **1303** returned to the host **1310** for the semantic search request. Further, in parallel, the non-volatile memory **1316** (e.g., DRAM) may be used for L2P table mapping, the data request **1305** may be sent to SSD storage device **1320**, processed (e.g., by SSD controller **1026/1126**) and then file data **1307** returned to the host **1310** for the file search/read/write/access request. Further, in some embodiments, where hyperscale or storage array is used, semantic searching and file data may be performed in parallel, and file data could be retrieved from conventional SSD with file replica from SSD storage device **1320**.

(117) Referring to FIG. **14**, a flowchart depicting a process **1400** for utilizing an example error correction method in a computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. The process **1400** can begin by storing raw file data in one or more memory structures (block **1410**). The process **1400** can store feature data corresponding to the raw file data in the corresponding one or more memory structures (block **1415**). The process **1400** can load feature data from at least one of the one or more memory structures, the feature data comprising at least one of: dense feature vectors or hyperdimensional vectors (block **1420**). The process **1400** can generate one or more databases from the feature data (block **1425**). The process **1400** can load the raw file data from at least one of the one or more memory structures (block

1430). The process **1400** can utilize an error correction algorithm for storing and accessing the feature data (block **1435**).

(118) Referring to FIG. **15**, a flowchart depicting a process **1500** for utilizing an example error correction method in a computational SSD system supporting semantic search in accordance with an embodiment of the disclosure. The process **1500** can begin by processing a query, using a machine learning model, for retrieving raw file data stored on one or more memory structures (block **1510**). The process **1500** can generate one or more databases from feature data stored in the one or more memory structures, the feature data comprises at least one of: dense feature vectors or hyperdimensional vectors (block **1515**). The process **1500** can load the one or more databases and raw file data from the one or more memory structures (block **1525**). The process **1500** can process the raw file data and one or more databases to obtain a response for the query (block **1525**).

(119) Information as herein shown and described in detail is fully capable of attaining the above-described object of the present disclosure, the presently preferred embodiment of the present disclosure, and is, thus, representative of the subject matter that is broadly contemplated by the present disclosure. The scope of the present disclosure fully encompasses other embodiments that might become obvious to those skilled in the art, and is to be limited, accordingly, by nothing other than the appended claims. Any reference to an element being made in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” All structural and functional equivalents to the elements of the above-described preferred embodiment and additional embodiments as regarded by those of ordinary skill in the art are hereby expressly incorporated by reference and are intended to be encompassed by the present claims.

(120) Moreover, no requirement exists for a system or method to address each and every problem sought to be resolved by the present disclosure, for solutions to such problems to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. Various changes and modifications in form, material, work-piece, and fabrication material detail can be made, without departing from the spirit and scope of the present disclosure, as set forth in the appended claims, as might be apparent to those of ordinary skill in the art, are also encompassed by the present disclosure.

Claims

1. A device comprising: a processor; a memory array comprising a plurality of memory devices; and a controller communicatively coupled to the memory array; one or more memory structures configured to store raw file data and feature data corresponding to the raw file data; wherein the feature data comprises at least one of: dense feature vectors based on error-free data and hyperdimensional vectors; wherein each of the one or more memory structures utilizes an error correction algorithm for storing and accessing the feature data; and wherein each of the one or more memory structures comprises a machine learning processing units configured to process the feature data.
2. The device of claim 1, wherein the dense feature vectors are stored in separate blocks from raw file data.
3. The device of claim 1, wherein the feature data is distributed within the one or more memory structures.
4. The device of claim 1, wherein the error correction algorithm is implemented as an on-die Error Correction Code (ECC) with each of the one or more memory structures having ECC.
5. The device of claim 1, wherein the feature data comprises hyperdimensional vectors, and at least two copies of the hyperdimensional vectors are stored on each of the one or more memory structures, a first copy of the hyperdimensional vectors stored without using an error correction algorithm, and a second copy stored using an error correction algorithm.

6. The device of claim 5, wherein the first copy of the hyperdimensional vector is provided to the machine learning processing unit, and the second copy of the hyperdimensional vector is provided for correcting errors in the first copy.
 7. A method comprising: storing raw file data in one or more memory structures; storing feature data corresponding to the raw file data in the corresponding one or more memory structures; loading feature data from at least one of the one or more memory structures; wherein the feature data comprises at least one of: dense feature vectors based on error-free data and hyperdimensional vectors; generating one or more databases from the feature data; loading the raw file data from at least one of the one or more memory structures; and utilizing an error correction algorithm for storing and accessing the feature data.
 8. The method of claim 7, wherein storing feature data comprises storing the dense feature vectors in separate blocks from the raw file data.
 9. The method of claim 7, further comprising distributing the feature data within the one or more memory structures.
 10. The method of claim 7, wherein the error correction algorithm is implemented as an on-die Error Correction Code (ECC) with each of the one or more memory structures having ECC.
 11. The method of claim 7, wherein the feature data comprises hyperdimensional vectors, and storing feature data comprises: storing at least two copies of the hyperdimensional vectors on each of the one or more memory structures, a first copy of the hyperdimensional vectors being stored without using an error correction algorithm, and a second copy being stored using an error correction algorithm.
 12. The method of claim 7, wherein each of the one or more memory structures includes a machine learning processing unit, and wherein a first copy of the hyperdimensional vector is provided to the machine learning processing unit, and a second copy of the hyperdimensional vector is provided for correcting errors in the first copy.
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